News Topic Classification Using Machine Learning and Transformer-Based Models

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Abstract

We compare three classical machine-learning baselines built on TF–IDF features against a fine-tuned `bert-base-uncased` encoder on the AG News topic-classification benchmark. All four systems are trained and evaluated under a single, controlled protocol: the official AG News training set is split 90/10 into 108,000 training and 12,000 development examples (stratified by label, seed `42`), and the official 7,600-example test set is held out until final evaluation. On the test set we report a Most Frequent baseline at 25.0% accuracy, Multinomial Naive Bayes at 90.24%, Logistic Regression at 91.80%, and BERT at 94.25% accuracy (94.24% macro-F1). Beyond the headline numbers, our main contribution is a paired error analysis of Logistic Regression and BERT on the same 7,600 test examples: 90.3% of inputs are classified correctly by both models, BERT corrects 4.0% that Logistic Regression misses, Logistic Regression corrects 1.5% that BERT misses, and 4.2% are missed by both. The bulk of BERT's net advantage concentrates on *Business* versus *Sci/Tech* confusions, where headlines share substantial surface vocabulary — a regime in which contextual representations noticeably outperform sparse n-gram features. Median headline length is roughly constant across BERT-correct, BERT-error, and Logistic-Regression-wrong/BERT-correct partitions, suggesting that the gain is not driven by length. We release all code, configurations, and predictions for reproducibility.[1]

1 Introduction

News topic classification is the task of assigning a short news item — typically a headline together with a leading paragraph — to one of a small number of pre-defined topical categories. The task is a workhorse of applied natural language processing: it underlies content recommendation, vertical search, automatic story routing, and many downstream content-analysis pipelines. From a research perspective, news classification is also a useful microcosm of supervised text classification more broadly, because the inputs are short, the labels are relatively unambiguous, and large, well-balanced public benchmarks are available.

For more than two decades, sparse bag-of-words and TF–IDF representations (Salton and Buckley, 1988) combined with linear or probabilistic classifiers have provided strong, well-understood baselines for topic classification. Multinomial Naive Bayes (McCallum and Nigam, 1998) is a particularly attractive starting point: it is essentially free to train, robust to high-dimensional sparse features, and surprisingly competitive on clean topical data. Regularised logistic regression on the same TF–IDF features pushes accuracy further at modest additional cost. With the rise of large pretrained masked-language-model encoders, in particular BERT (Devlin et al., 2019), the empirical state of the art on most public text classification benchmarks — including AG News (Zhang et al., 2015) — has shifted decisively toward fine-tuned contextual encoders. The question that motivates this work is therefore not whether transformers help, but *by how much, and where exactly*.

Concretely, we ask:

> *How much does a transformer-based model like BERT improve news topic classification compared with traditional TF–IDF-based classifiers, and on which kinds of examples does that improvement appear?*

We answer this question on the AG News bench-

mark by training and evaluating four systems under a single, controlled protocol: a Most Frequent baseline, Multinomial Naive Bayes on TF–IDF unigram and bigram features, Logistic Regression on the same features, and a fine-tuned `bert-base-uncased` encoder. Beyond the standard accuracy/precision/recall/F1 comparison, we conduct a paired error analysis on the same 7,600 test examples to localise where BERT's gains actually come from.

Contributions. The contributions of this paper are:

1. A controlled four-model comparison (Most Frequent, Naive Bayes, Logistic Regression, BERT) on AG News under a single 90/10 stratified train/dev split with the official held-out test set.
2. A *paired* test-set error analysis of Logistic Regression versus BERT, decomposing the 7,600 test examples into *both correct*, *LR wrong / BERT correct*, *BERT wrong / LR correct*, and *both wrong*.
3. A confusion-pair-level analysis showing that BERT's net gains concentrate on *Business* vs. *Sci/Tech* confusions, with qualitative examples where surface keywords are misleading.
4. A length-based sanity check showing that BERT's improvement is not explained by headline length.
5. A fully reproducible code release with deterministic seeding, saved experiment configurations, per-model prediction CSVs, and per-model confusion matrices.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the AG News dataset and our split. Section 4 introduces the four model families. Section 5 details the experimental protocol. Section 6 reports headline accuracy/F1 results and per-class behaviour. Section 7 presents the paired error analysis. Section 8 discusses implications, Section 9 states limitations, and Section 10 concludes.

2 Related Work

Classical text classification. The combination of bag-of-words representations, TF–IDF weighting (Salton and Buckley, 1988), and a linear or probabilistic classifier has been a default baseline for topic classification since at least the late 1990s. Multinomial Naive Bayes (McCallum and Nigam, 1998) models class- conditional word distributions under a conditional-independence assumption and remains competitive on short, topically coherent inputs. Regularised logistic regression replaces the generative assumption with a discriminative one, and consistently outperforms Naive Bayes on high-dimensional sparse text features in our setting and in prior work.

Neural text classification before transformers. Convolutional networks operating on pretrained word embeddings (Kim, 2014) provided the first widely-replicated demonstration that neural architectures could exceed strong sparse linear baselines on sentence-level classification. Hierarchical attention networks (Yang et al., 2016) extended this idea to document-level classification by attending over both words and sentences. Efficient bag-of-tricks classifiers such as fastText (Joulin et al., 2017) showed that, with careful engineering, simple linear models on subword embeddings can match the accuracy of much heavier architectures on topic classification in a fraction of the training time.

Transformer-based classification. The Transformer architecture (Vaswani et al., 2017) and the masked-language- modelling pretraining recipe of BERT (Devlin et al., 2019) re-set the state of the art on most supervised classification benchmarks. For sequence classification, BERT prepends a special `[CLS]` token whose final hidden state is fed into a single linear classification head. The model is then fine-tuned end-to-end on the downstream supervised data. On AG News specifically, fine-tuned transformer encoders are now the dominant approach (Devlin et al., 2019; Zhang et al., 2015), although well-tuned linear baselines on TF–IDF features remain remarkably competitive on this corpus, which motivates a careful comparative study rather than a foregone conclusion.

This work. We deliberately keep the model space small (one Most Frequent baseline, two TF–IDF classifiers, one fine-tuned BERT) and the experimental protocol uniform across all four systems, in order to focus the empirical analysis on *where* transformer-based representations help relative to sparse n-gram features. Our paired error-analysis methodology is closest in spirit to comparative-evaluation work in the NLP literature where the same test examples are scored by multiple systems and the disagreements are inspected qualitatively and quantitatively.

3 Dataset

We use the AG News topic-classification corpus (Zhang et al., 2015) as distributed through the Hug-

Table 1: Dataset splits. AG News has four balanced classes (*World*, *Sports*, *Business*, *Sci/Tech*). The 108k/12k partition of the training set is stratified by label with seed `42`.

Split	#Examples
Train	108,000
Dev	12,000
Test	7,600
Total	127,600

ging Face `datasets` library (`ag_news` configuration). Each example is a short news item consisting of a headline and a leading description, labelled with one of four topical classes: *World*, *Sports*, *Business*, and *Sci/Tech* (label IDs 0–3 respectively). The dataset is balanced: each class has 30,000 training examples and 1,900 test examples.

Splits. The official AG News release contains a 120,000-example training set and a 7,600-example test set. We further partition the training set into a 108,000-example training portion and a 12,000-example development portion using a stratified `train_test_split` from the `datasets` library (test fraction 0.1, seed `42`, stratified by label), which maintains the four-way class balance in both halves. The development set is used for sanity checks and to monitor BERT training; the official 7,600-example test set is held out and used only for the final reported numbers. Table 1 summarises the splits.

Input format. Each AG News example concatenates the headline and the leading description into a single text field. Inputs are short: in our test split, the median word count (whitespace-separated tokens) is 37, the mean is 37.7, and the empirical range is 11–137. This short-input regime favours efficient classifiers and limits the benefit of long-context architectures.

4 Methodology

We compare four systems: a Most Frequent baseline, Multinomial Naive Bayes, Logistic Regression, and a fine-tuned BERT encoder. The first three operate on a shared TF–IDF feature representation; the fourth operates directly on tokenised text.

4.1 Shared Notation

Let $x \in \mathcal{V}^*$ denote an input headline-plus-description, $\phi(x)$ a feature representation, and $y \in \{0, 1, 2, 3\}$ the topic label corresponding to *World*, *Sports*, *Business*, and *Sci/Tech* respectively.

4.2 Most Frequent Baseline

The Most Frequent baseline predicts a single label — the modal class on the training set — for every test example. AG News is balanced, so this baseline is expected to score near 25% accuracy. We include it to provide a hard lower bound and to verify that all subsequent models clear the trivial floor:

$$\hat{y} \;=\; \arg\max_{y} \#\{i : y_i = y\}. \tag{1}$$

4.3 TF–IDF Feature Extraction

For the two classical learners we extract sparse TF–IDF features with scikit-learn's `TfidfVectorizer` (Pedregosa et al., 2011) using the following configuration:

- lower-casing,
- English stop-word removal (scikit-learn built-in list),
- unigram and bigram features (`ngram_range` $= (1, 2)$),
- vocabulary capped at the top 20,000 features by document frequency.

The vectoriser is fit on the 108,000 training examples only, and then applied unchanged to the development and test splits, so no information from the held-out evaluation data is leaked into the features.

4.4 Multinomial Naive Bayes

Multinomial Naive Bayes (McCallum and Nigam, 1998) models class-conditional word distributions under a conditional-independence assumption:

$$\hat{y} \;=\; \arg\max_{y} p(y) \prod_{w \in \phi(x)} p(w \mid y), \tag{2}$$

where $p(w \mid y)$ is estimated from training-set counts with Laplace smoothing. We use scikit-learn's default `MultinomialNB` on the 20,000-feature TF–IDF representation; no hyperparameters are tuned.

4.5 Logistic Regression

We train a multinomial logistic regression with ℓ_2 regularisation on the same TF–IDF features:

$$\min_{W} \frac{1}{N} \sum_{i=1}^{N} \ell\big(W^\top \phi(x_i),\, y_i\big) + \lambda \, \|W\|_2^2. \tag{3}$$

We use scikit-learn's logistic regression solver (`LogisticRegression`) with default ℓ_2 regularisation, `max_iter` set to 2000, and `random_state`

Table 2: BERT fine-tuning configuration. All other settings follow the Hugging Face `Trainer` defaults (`transformers`-style logging and evaluation, mixed precision disabled). `load_best_model_at_end=True` restores the checkpoint with the best dev-set evaluation loss.

Hyperparameter	Value
Base model	`bert-base-uncased`
Number of labels	4
Max sequence length	128
Optimiser	AdamW
Learning rate	2×10^{-5}
Batch size (train/eval)	8 / 8
Number of epochs	2
Weight decay	0.01
Random seed	42
Eval/save strategy	per epoch

set to 42. This model is the strongest non-neural baseline in our pipeline and serves as the focal point of the paired error analysis in Section 7.

4.6 BERT Fine-Tuning

We fine-tune `bert-base-uncased` (Devlin et al., 2019) (12 transformer layers, 768 hidden units, 12 attention heads, ∼110M parameters) for four-way classification. Inputs are tokenised with the model's WordPiece tokenizer and truncated to a maximum length of 128 sub-word tokens. The pretrained encoder is loaded with the Hugging Face `transformers` library (Wolf et al., 2020) via `AutoModelForSequenceClassification`, which appends a single randomly-initialised linear classification head over the final hidden state of the `[CLS]` token:

$$p(y \mid x) \;=\; \mathrm{softmax}\big(W\, h_{\texttt{[CLS]}} + b\big). \qquad (4)$$

The encoder and the classification head are fine-tuned jointly on the 108,000-example training split using the AdamW optimiser (Loshchilov and Hutter, 2019) with the hyperparameters listed in Table 2. Due to compute and time constraints, we run a single fine-tuning configuration rather than a hyperparameter search.

5 Experimental Setup

5.1 Evaluation Metrics

We report four metrics on the official 7,600-example test set: overall accuracy, and macro-averaged precision, recall, and F1. Macro averaging gives equal weight to each of the four classes regardless of support, which on AG News differs negligibly from micro averaging because the test set is balanced (1,900 examples per class). Per- class precision, recall, and F1 are computed from the corresponding confusion matrix.

Table 3: Test-set results on AG News ($N = 7{,}600$). Best score per column in **bold**. Precision, recall, and F1 are macro-averaged.

Model	Acc	Prec	Rec	F1
Most Frequent	0.2500	0.0625	0.2500	0.1000
Naive Bayes	0.9024	0.9019	0.9024	0.9019
Logistic Reg.	0.9180	0.9178	0.9180	0.9178
BERT	**0.9425**	**0.9424**	**0.9425**	**0.9424**

5.2 Reproducibility

All randomised steps use seed `42`. Specifically: (i) the 90/10 train/dev split is performed with `seed=42` and stratified by label; (ii) the Logistic Regression solver is invoked with `random_state=42`; (iii) the Hugging Face `Trainer` receives `seed=42`; and (iv) NumPy and Python's `random` are seeded with `42` at the top of `utils.py`. Each training script also emits a JSON configuration file under `results/csv/` that records the dataset split sizes, hyperparameter values, and the precise versions of `numpy`, `pandas`, `scikit-learn`, `datasets`, `transformers`, `torch`, `accelerate`, and `matplotlib` that were used. Predictions on the test set are serialised to CSV (`lr_test_predictions.csv` and `bert_test_predictions.csv`) so that the paired error analysis of Section 7 can be re-run without retraining.

5.3 Software and Hardware

The implementation uses Python 3.10, scikit-learn 1.5.2 for the classical pipelines, and PyTorch 2.4.1 with Hugging Face `transformers` 5.6.0 and `accelerate` 1.13.0 for the BERT encoder. The classical pipelines run on CPU; BERT is fine-tuned on a single consumer-grade GPU. A reproducible `requirements.txt` pinning all package versions is included in the repository.

6 Results

6.1 Main Results

Table 3 reports the four standard metrics on the AG News test set; Figure 1 shows the same numbers visually.

Three observations emerge.

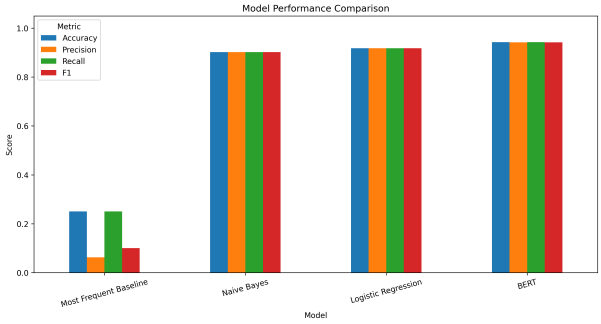

Figure 1: Test-set accuracy, precision, recall, and macro-F1 for the four models on AG News. The Most Frequent baseline collapses to a single class and so its precision is far below its recall; Naive Bayes, Logistic Regression, and BERT have nearly identical precision and recall because the test set is balanced.

All non-trivial models clear the floor by a wide margin. The Most Frequent baseline scores 0.25 accuracy, exactly as expected on a balanced four-class corpus. Naive Bayes, Logistic Regression, and BERT all exceed 0.90 accuracy.

Logistic Regression is the strongest classical baseline. Logistic Regression reaches 91.80% test accuracy and 91.78% macro-F1, beating Multinomial Naive Bayes by 1.56 accuracy points and 1.59 F1 points. This reproduces the long-standing empirical finding that linear discriminative models are well matched to high- dimensional sparse text representations and tend to outperform generative bag-of-words models on TF–IDF features.

BERT achieves the best overall performance. The fine-tuned `bert-base-uncased` encoder reaches 94.25% accuracy and 94.24% macro-F1, an improvement of 2.45 accuracy points and 2.46 F1 points over Logistic Regression, and 4.01 accuracy points over Naive Bayes. The improvement over the strongest non-neural baseline is meaningful but not large, suggesting that for clean, balanced, short-text topical news data, sparse n-gram features already carry most of the signal.

6.2 Per-Class Confusion Structure

Figures 2 and 3 show the per-class confusion matrices for Logistic Regression and BERT respectively, and Table 4 reports the per-row diagonal counts on the test set. Each test class has 1,900 ground-truth examples, so the diagonal entries can be interpreted directly as per-class recall counts.

Three patterns are consistent across all non-trivial models. First, *Sports* is by far the easiest class: Logistic Regression correctly classifies 1,864/1,900 (98.1% recall) and BERT 1,883/1,900 (99.1% recall). Sports headlines contain a stable, distinctive vocabulary (athlete names, league acronyms, match-action verbs) that bag-of-words features capture well. Second, *Business* and *Sci/Tech* are the hardest pair: in the Logistic Regression matrix, 147 Business examples are predicted as Sci/Tech and 118 Sci/Tech examples are predicted as Business; the analogous BERT counts are 139 and 86. Third, BERT improves the most absolute recall on *World* (+72 correct) and *Sci/Tech* (+51 correct) relative to Logistic Regression, followed by *Business* (+44); the *Sports* class is already close to ceiling for both models.

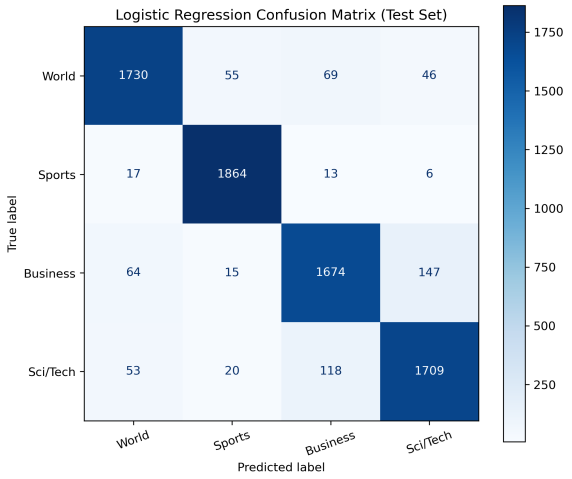

Figure 2: Logistic Regression confusion matrix on the AG News test set ($N = 7{,}600$, 1,900 examples per class).

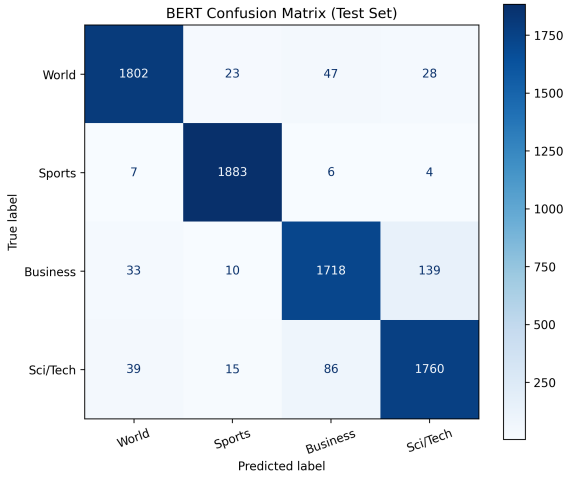

Figure 3: BERT confusion matrix on the AG News test set ($N = 7{,}600$, 1,900 examples per class).

Table 4: Per-class diagonals from the test-set confusion matrices. Each class has 1,900 ground-truth examples, so the values are correctly classified counts; per-class recall is the value divided by 1,900.

Model	World	Sports	Business	Sci/Tech
Most Freq.	0	0	1900	0
Naive Bayes	1705	1866	1605	1682
Logistic Reg.	1730	1864	1674	1709
BERT	1802	1883	1718	1760

Table 5: Paired test-set overlap of Logistic Regression and BERT ($N = 7{,}600$). BERT corrects 301 examples that LR misses, while LR corrects 115 that BERT misses — a net gain of 186 examples (≈ 2.45 accuracy points), accounting for essentially the entire accuracy gap.

Case	Count	Share
Both correct	6,862	90.3%
LR wrong / BERT correct	301	4.0%
BERT wrong / LR correct	115	1.5%
Both wrong	322	4.2%
Total	7,600	100.0%

7 Error Analysis: BERT vs. Logistic Regression

The aggregate-metrics view in Section 6 does not show *which* examples each model gets right. To localise BERT's advantage, we run a paired analysis on the same 7,600 test examples by joining the per-example predictions of Logistic Regression and BERT (saved as `lr_test_predictions.csv` and `bert_test_predictions.csv`) on the input text and the gold label.

7.1 Overlap Decomposition

We partition the test set into four mutually exclusive cases: *both correct*, *LR wrong / BERT correct*, *BERT wrong / LR correct*, and *both wrong*. Table 5 and Figure 4 report the counts and shares.

The dominant outcome is *both correct*: 6,862 test examples (90.3%) are classified correctly by both models. Only the 4,0% *LR wrong / BERT correct* stripe and the 1.5% *BERT wrong / LR correct* stripe contribute to BERT's net accuracy gain. The arithmetic confirms the headline numbers: BERT's net advantage of $301 - 115 = 186$ examples corresponds to $186/7600 = 2.45$ accuracy points, matching the $0.9425 - 0.9180$ gap in Table 3 to two decimal places.

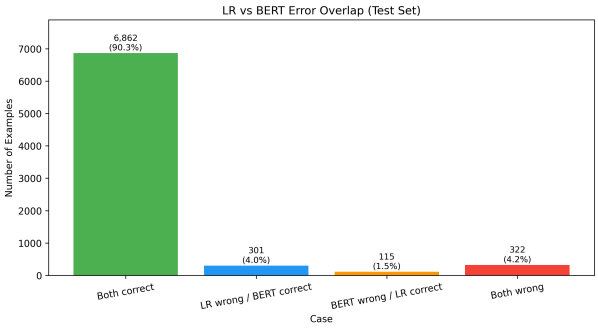

Figure 4: Paired error overlap of Logistic Regression and BERT on the AG News test set. The two non-symmetric bars (*LR wrong / BERT correct* = 301 vs. *BERT wrong / LR correct* = 115) show that BERT's net advantage is 186 examples, which is essentially the entire accuracy gap reported in Table 3.

Table 6: Most frequent Logistic Regression confusion pairs in the *LR wrong / BERT correct* stripe ($n = 301$). Format: *gold* $\rightarrow$ *predicted*.

LR confusion pair	Count
Sci/Tech $\rightarrow$ Business	56
Business $\rightarrow$ Sci/Tech	44
Business $\rightarrow$ World	40
World $\rightarrow$ Sports	38
World $\rightarrow$ Business	31
World $\rightarrow$ Sci/Tech	25
Sci/Tech $\rightarrow$ World	23
Sports $\rightarrow$ World	14
Sci/Tech $\rightarrow$ Sports	10
Business $\rightarrow$ Sports	9
Sports $\rightarrow$ Business	7
Sports $\rightarrow$ Sci/Tech	4

7.2 Where BERT Helps

Examples where Logistic Regression is wrong and BERT is correct ($n = 301$) are dominated by confusions between topic pairs that share substantial surface vocabulary. Table 6 lists the most frequent Logistic Regression confusion pairs in this stripe.

The two largest groups, *Sci/Tech*$\rightarrow$*Business* (56) and *Business*$\rightarrow$*Sci/Tech* (44), together account for 33% of the BERT-corrected examples. To avoid presenting whatever happens to sit at the top of `lr_wrong_bert_correct.csv`, we draw a *stratified* sample of one illustrative case from each of four distinct Logistic Regression confusion pairs that are well represented in the BERT-corrected stripe:

- **[Sci/Tech$\rightarrow$Business]** *"IBM to hire even more new workers. By the end of the year, the computing giant plans to have its biggest headcount since 1991."* Gold: *Sci/Tech*. Logistic Regression

Table 7: Both-wrong confusion pairs on the AG News test set, computed using BERT's predicted class on each both-wrong example. Pairs are symmetrised across direction.

Both-wrong pair	Count
Business ↔ Sci/Tech	164
World ↔ Business	62
Sci/Tech ↔ World	50
Sports ↔ World	22
Sci/Tech ↔ Sports	14
Business ↔ Sports	10

predicts *Business*, plausibly responding to the keywords *hire*, *workers*, and *headcount*; BERT predicts *Sci/Tech*, which is consistent with *IBM*'s identity as a technology company in this corpus.

- **[Business→Sci/Tech]** *"Justices to debate mail-order wine…"* Gold: *Business*. Logistic Regression predicts *Sci/Tech*, while BERT predicts *Business*.
- **[World→Sports]** *"Live: Olympics day four. Richard Faulds and Stephen Parry are going for gold for Great Britain in Athens."* Gold: *World*. Logistic Regression predicts *Sports*, presumably responding to the *Olympics*/*gold*/*Athens* cluster; BERT predicts *World*, which is the AG News convention for Olympics-as-international-news framing.
- **[World→Business]** *"India's Tata expands regional footprint via NatSteel buyout."* Gold: *World*. Logistic Regression predicts *Business* on the keyword *buyout*; BERT predicts *World*.

A common thread runs through these cases: Logistic Regression assigns weight to individual keywords (*hire*, *workers*, *Olympics*, *buyout*) that strongly co-occur with one class but in fact appear in headlines whose overall topic is a different class. BERT's contextual representations down-weight those keywords in the presence of disambiguating context (the headline-level identity of *IBM*, the international-news framing around *Olympics*, and so on). This is the same surface-vs-context distinction that drives the canonical *Apple* (company) vs. *apple* (fruit) example from the lexical-semantics literature.

7.3 Where BERT Still Struggles

The *both wrong* stripe ($n = 322$, 4.2% of the test set) is also concentrated on the same hard topic pairs. Table 7 reports counts for the dominant pairs (symmetrised in the *Topic A* ↔ *Topic B* direction).

Half of the both-wrong residual ($164/322 = 51\%$) is the *Business* ↔ *Sci/Tech* pair. Inspecting these examples manually, many are genuinely ambiguous from the headline alone — a story about a major technology firm's earnings or acquisition can be reasonably labelled either way, and AG News's ground-truth labels are not always consistent across such stories. This suggests that any further accuracy improvement on AG News will have to come either from external knowledge (entity-type lookups, ontologies) or from longer context than the 128 subword tokens used here, rather than from a stronger encoder alone.

7.4 Headline-Length Sanity Check

A natural alternative explanation for BERT's gain would be that BERT disproportionately helps on either very long or very short headlines. We test this by comparing the median whitespace-token count on four test-set partitions:

- the full test set,
- BERT-correct examples,
- BERT-error examples,
- LR-wrong / BERT-correct examples.

The median token count is approximately 36 in all four partitions (37 on the full test set, 37 on BERT-correct, 36 on BERT-error, and 35 on LR-wrong/BERT-correct). The differences are within a single word, so headline length is essentially constant across the correctness partitions. We conclude that BERT's net advantage is *not* explained by length, and is more plausibly attributable to its contextual handling of polysemous keywords, as documented in Section 7.2.

8 Discussion

The transformer advantage is real but bounded on AG News. BERT improves test accuracy by 2.45 points over Logistic Regression and by 4.01 points over Naive Bayes. While these gains are statistically meaningful on 7,600 test examples, they are smaller than the gaps frequently reported on harder language-understanding benchmarks such as natural language inference or question answering (Devlin et al., 2019; Vaswani et al., 2017). The intuition is straightforward: AG News is a clean, balanced, short-input topical-news corpus on which class-conditional vocabulary already carries most of the signal, so the marginal benefit of a contextual encoder is constrained.

The gain concentrates on lexically-overlapping topics. The paired error analysis localises BERT's net gain almost entirely on *Business*-versus-*Sci/Tech*, *World*-versus- *Business*, and Olympics-style *World*-versus- *Sports* confusions — precisely the topic pairs where surface keywords are misleading. On the easy class (*Sports*), Logistic Regression already operates near ceiling, and BERT can only add a small absolute improvement.

Cost matters in practice. Logistic Regression on TF–IDF reaches 91.8% accuracy and trains in seconds on a CPU; BERT reaches 94.25% accuracy but requires GPU fine-tuning and many orders of magnitude more parameters and floating-point operations. For a deployment in which a ~ 2.5 percentage-point accuracy drop is acceptable, the linear baseline is a defensible operating point. For a deployment in which accuracy is the top priority and a GPU budget is available, BERT is the dominant choice. Our paired analysis sharpens the trade-off by showing *which* test examples actually move when the classifier is upgraded.

9 Limitations

We report results from a single run per model with a fixed seed and have not estimated variance across multiple random seeds; the 0.25%–2.5% accuracy gaps we report are therefore lower bounds on the variance-aware comparison one would need for tight statistical claims. We did not perform a hyperparameter search for BERT (single configuration: 2 epochs, batch size 8, learning rate 2×10^{-5}, max length 128). AG News is a clean, balanced benchmark and our evaluation is in-distribution; we have not tested the trained models on a different news source, time period, or messier real-world inputs. The inputs are short headlines and leading paragraphs (median 37 whitespace tokens), so the conclusions may not transfer to full-length articles. Finally, we report accuracy/precision/recall/F1 but not calibration: the BERT classification head is fine-tuned with cross- entropy and is known to be miscalibrated on out-of-distribution data even when accuracy is high, and we have not analysed this here.

10 Conclusion and Future Work

We presented a controlled four-model comparison on the AG News topic- classification benchmark and a paired error analysis of Logistic Regression versus a fine-tuned BERT encoder. Our headline numbers reproduce the broad intuition of the literature — BERT is best, with 94.25% test accuracy, and a TF–IDF Logistic Regression is a strong non-neural baseline at 91.80% accuracy. The paired analysis sharpens this comparison by showing that BERT's net advantage of 186 examples accounts for essentially the entire accuracy gap, and that this advantage concentrates on *Business*-versus-*Sci/Tech* and other lexically-overlapping topic confusions. A length-based sanity check shows that BERT's gain is not driven by headline length.

Several extensions follow naturally. First, comparing BERT to lighter distilled or alternative transformer encoders (DistilBERT, RoBERTa, DeBERTa) under the same protocol would clarify how much of the accuracy gain is specific to `bert-base-uncased` versus generic to pretrained transformers. Second, multiple random seeds and a hyperparameter sweep would let us report variance-aware accuracy gaps. Third, an out-of-distribution evaluation on a more recent or messier news corpus would test whether BERT's advantage is robust to domain shift. Fourth, a confidence-and-calibration analysis on the four-way softmax outputs would complement the accuracy/F1 picture with deployment-relevant uncertainty information. Fifth, a layer-wise probing analysis — training a frozen-features linear classifier on the hidden states at each of BERT's 12 transformer layers (Tenney et al., 2019) — would localise which layers carry the topic- classification signal on AG News, and could motivate cheaper fine-tuning strategies that update only a subset of layers or use parameter-efficient adapters. Sixth, the paired error-analysis methodology used here generalises directly to other tasks where two competing classifiers can be evaluated on the same test set.

Author Contributions

All four authors contributed jointly to the design of the experiments, the implementation of the data-loading and evaluation pipeline, the training and evaluation of the four models, the paired error analysis, and the preparation of this manuscript. Specific responsibilities and fine-grained breakdowns are recorded internally and are available on request.

Reproducibility Statement

Source code, training and analysis scripts, per-model test predictions, confusion matrices, overlap-analysis output, and pinned dependency versions

are publicly available in the project repository.[2] The repository contains the six Python source files (`baseline_models`, `bert_model`, `error_analysis`, `overlap_analysis`, `plot`, and `utils`), an automatically populated `results/` directory, and a `requirements.txt` with pinned package versions. The training scripts also write JSON experiment-configuration files to `results/csv/` that record dataset split sizes, hyperparameter values, and the precise package versions used.

[2] github.com/kechavious/News-Topic-Classification-Using-Machine-Learning-and-Transformer-Based-Models

References

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pp. 4171–4186, 2019.

Armand Joulin, Edouard Grave, Piotr Bojanowski, and Tomas Mikolov. Bag of tricks for efficient text classification. In *Proceedings of EACL (Volume 2: Short Papers)*, pp. 427–431, 2017.

Yoon Kim. Convolutional neural networks for sentence classification. In *Proceedings of EMNLP*, pp. 1746–1751, 2014.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *Proceedings of ICLR*, 2019.

Andrew McCallum and Kamal Nigam. A comparison of event models for naive Bayes text classification. In *AAAI Workshop on Learning for Text Categorization*, 1998.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12(85):2825–2830, 2011.

Gerard Salton and Christopher Buckley. Term-weighting approaches in automatic text retrieval. *Information Processing & Management*, 24(5):513–523, 1988.

Ian Tenney, Dipanjan Das, and Ellie Pavlick. BERT rediscovers the classical NLP pipeline. In *Proceedings of ACL*, pp. 4593–4601, 2019.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, pp. 5998–6008, 2017.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP: System Demonstrations*, pp. 38–45, 2020.

Zichao Yang, Diyi Yang, Chris Dyer, Xiaodong He, Alex Smola, and Eduard Hovy. Hierarchical attention networks for document classification. In *Proceedings of NAACL-HLT*, pp. 1480–1489, 2016.

Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*, pp. 649–657, 2015.